

1. Visible units: v

The visible units represent the data that we observe.

For an MNIST image:

- Each visible unit may represent one pixel.
- A binary visible unit is usually either 0 or 1.
- 0 might represent an inactive or dark pixel.
- 1 might represent an active or bright pixel.

Input image pixels

$v_1 \ v_2 \ v_3 \ v_4 \ \dots \ v_{784}$

They are called **visible** because their values come directly from the data.

2. Hidden units: h

The hidden units are not given directly by the dataset. They learn patterns in the visible data.

For MNIST, hidden units might respond to patterns such as:

- a vertical line
- a curve
- a loop
- the top part of a digit
- combinations of pixels that frequently occur together

Hidden features

$h_1 \ h_2 \ h_3 \ \dots \ h_n$

Like the visible units, traditional RBM hidden units are often binary: on or off.

3. Learned weights

Each visible unit is connected to each hidden unit through a learned weight:

w_{ij}

For example:

Visible pixel v_i — w_{ij} — Hidden feature h_j

The weight tells the model how compatible those two units are.

- A large positive weight encourages both units to be active together.
- A large negative weight discourages them from being active together.
- A weight near zero means the relationship is weak.

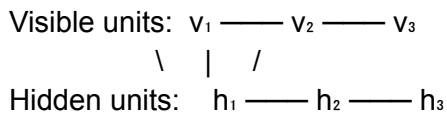
The model also learns biases for visible and hidden units.

4. What is restricted?

A full Boltzmann machine can have:

- visible-to-visible connections
- hidden-to-hidden connections
- visible-to-hidden connections

Full Boltzmann machine

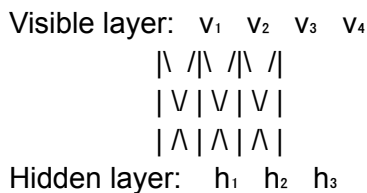


This creates many dependencies and makes sampling difficult.

A **restricted Boltzmann machine** removes:

- all visible-to-visible connections
- all hidden-to-hidden connections

Only visible-to-hidden connections remain:



That is why it is called **restricted**.

5. What does bipartite graph mean?

A bipartite graph has two separate groups of nodes:

Group 1: Visible units

Group 2: Hidden units

Connections are allowed only between the two groups.

There are no connections within either group.

So:

Visible ↔ Hidden	allowed
Visible ↔ Visible	not allowed
Hidden ↔ Hidden	not allowed

6. Why does this make alternating updates possible?

Because there are no connections between hidden units, once the visible units are known, every hidden unit can be computed independently.

Visible image v



Sample all hidden units h

Then, because there are no connections between visible units, once the hidden units are known, every visible unit can also be computed independently.

Hidden features h



Sample all visible units v

The model alternates:

$v \rightarrow h \rightarrow v \rightarrow h \rightarrow v \rightarrow h$

This is Gibbs sampling.

For example:

Original image



Sample hidden features



Reconstruct visible pixels



Sample hidden features again



Reconstruct the image again

Each alternation moves the network toward a configuration that fits its learned energy function.

MNIST interpretation

Visible units

Pixel values of a handwritten digit



Learned weights



Hidden units

Stroke, curve, loop, and shape patterns

During reconstruction:

Hidden patterns



Generate likely pixel values



Reconstructed digit

The key idea is:

Removing connections within each layer makes the units in one layer independent when the other layer is known. That allows the model to update an entire layer at once and alternate between the visible and hidden layers.